Calculation of Free energy of hydration of Ethanol

For NVT Batch (0-4)

```
for i in {0..4}; do
   echo "========================================="
   echo "RUNNING NVT STATE $i (GPU Accelerated)"
   echo "========================================="
   gmx grompp -f nvt_$i.mdp -c em_$i.gro -p topol.top -o nvt_$i.tpr -maxwarn 2
   gmx mdrun -deffnm nvt_$i -nb gpu -pme gpu
   echo "State $i complete. Cooling down fanless chassis for 5 minutes (300s)..."
   sleep 300
done
```

Note: for the mdp files I'm taking sc-alpha = 1 instead of the standard 0.5 since using 0.5 has caused issues in the previous project (methane in water) where the system was blowing up so increasing the soft core potential reduces the chances of the system blowing up for certain lambda values (issues caused for lambda values 0.25 and 0.8)
And pme_order = 4 instead of 6 since gpu acceleration for pme isn't supported for anything other than pme_order = 4. Accuracy isn't affected that much since the system is very small



```
Command line:
  gmx mdrun -deffnm nvt_0 -nb gpu -pme gpu

Reading file nvt_0.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      666.629      111.106      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       77.765        0.309        2.222            2.920

GROMACS reminds you: "It Wouldn't Hurt to Wipe Once In a While" (Beavis and Butthead)

State 0 complete. Cooling down fanless chassis for 5 minutes (300s)...
```

```
Command line:
  gmx mdrun -deffnm nvt_1 -nb gpu -pme gpu

Reading file nvt_1.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      638.117      106.354      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       81.240        0.295        2.127            3.051

GROMACS reminds you: "Predictions can be very difficult - especially about the future." (Niels Bohr)

State 1 complete. Cooling down fanless chassis for 5 minutes (300s)...
========================================
```

```
Command line:
  gmx mdrun -deffnm nvt_2 -nb gpu -pme gpu

Reading file nvt_2.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      831.376      138.563      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       62.355        0.385        2.771            2.342

GROMACS reminds you: "Do You Have a Mind of Your Own ?" (Garbage)

State 2 complete. Cooling down fanless chassis for 5 minutes (300s)...
========================================
```

```
Command line:
  gmx mdrun -deffnm nvt_3 -nb gpu -pme gpu

Reading file nvt_3.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      639.596      106.600      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       81.052        0.296        2.132            3.044

GROMACS reminds you: "Can't You Make This Thing Go Faster ?" (Black Crowes)

State 3 complete. Cooling down fanless chassis for 5 minutes (300s)...
========================================
```

```
Command line:
  gmx mdrun -deffnm nvt_4 -nb gpu -pme gpu

Reading file nvt_4.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      637.322      106.221      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       81.341        0.295        2.124           3.055

GROMACS reminds you: "Forget this world and all its troubles and, if possible, it's multitudinous Charlatans – everything, in short, but the Enchantress of Numbers" (Ada Lov

State 4 complete. Cooling down fanless chassis for 5 minutes (300s)...
```

```
for i in {5..9}; do
   echo "========================================="
   echo "RUNNING NVT STATE $i (GPU Accelerated)"
   echo "========================================="
   gmx grompp -f nvt_$i.mdp -c em_$i.gro -p topol.top -o nvt_$i.tpr -maxwarn 2
   gmx mdrun -deffnm nvt_$i -nb gpu -pme gpu
   echo "State $i complete. Cooling down fanless chassis for 5 minutes (300s)..."
   sleep 300
done
```

```
  gmx mdrun -deffnm nvt_5 -nb gpu -pme gpu

Reading file nvt_5.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      778.458      129.744      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       66.594        0.360        2.595           2.501

GROMACS reminds you: "By denying scientific principles, one may maintain any paradox." (Galileo Galilei)

State 5 complete. Cooling down fanless chassis for 5 minutes (300s)...
```

```
  gmx mdrun -deffnm nvt_6 -nb gpu -pme gpu

Reading file nvt_6.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

Starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      974.360      162.395      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       53.205        0.451        3.248            1.998

GROMACS reminds you: "Optimist: The glass is 1/2 full. Pessimist: The glass is 1/2 empty. Excel: The glass is Janu
d." (John Feminella)

State 6 complete. Cooling down fanless chassis for 5 minutes (300s)...
================================================
```

```
Command line:
  gmx mdrun -deffnm nvt_7 -nb gpu -pme gpu

Reading file nvt_7.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

Starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      873.687      145.615      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       59.336        0.404        2.912            2.228

GROMACS reminds you: "I was elected to lead, not to read" (President A. Schwarzenegger)

State 7 complete. Cooling down fanless chassis for 5 minutes (300s)...
================================================
```

```
  gmx mdrun -deffnm nvt_8 -nb gpu -pme gpu

Reading file nvt_8.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     1558.259      259.711      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       33.268        0.721        5.194            1.249

GROMACS reminds you: "Before we work on artificial intelligence why don't we do something about natural stupidity?" (Ste
ve Polyak)

State 8 complete. Cooling down fanless chassis for 5 minutes (300s)...
=======================================
RUNNING NVT STATE 9 (GPU Accelerated)
=======================================
Data prefix:  /opt/gromacs
Working dir:  /Users/saichaitanyajagarlapudi/GROMACS_Workspace/free energy of hydration of ethanol in water calculation
Command line:
  gmx mdrun -deffnm nvt_9 -nb gpu -pme gpu

Reading file nvt_9.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      934.483      155.748      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       55.475        0.433        3.115            2.083

GROMACS reminds you: "Three Little Fonzies" (Pulp Fiction)

State 9 complete. Cooling down fanless chassis for 5 minutes (300s)...
```

```
for i in {10..14}; do
  echo "========================================="
  echo "RUNNING NVT STATE $i (GPU Accelerated)"
  echo "========================================="
  gmx grompp -f nvt_$i.mdp -c em_$i.gro -p topol.top -o nvt_$i.tpr -maxwarn 2
```

```
    gmx mdrun -deffnm nvt_$i -nb gpu -pme gpu
    echo "State $i complete. Cooling down fanless chassis for 5 minutes (300s)..."
    sleep 300
done
```

```
   gmx mdrun -deffnm nvt_10 -nb gpu -pme gpu


Back Off! I just backed up nvt_10.log to ./#nvt_10.log.1#
Reading file nvt_10.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads


Back Off! I just backed up nvt_10.xtc to ./#nvt_10.xtc.1#

Back Off! I just backed up nvt_10.edr to ./#nvt_10.edr.1#

Back Off! I just backed up nvt_10.xvg to ./#nvt_10.xvg.1#
starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      672.977      112.164      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       77.032        0.312        2.243            2.893

GROMACS reminds you: "BioBeat is Not Available In Regular Shops" (P.J. Meulenhoff)
```

```
Data prefix:  /opt/gromacs
Working dir:  /Users/saichaitanyajagarlapudi/GROMACS_Workspace/free energy of hydration of ethanol in water calculation
Command line:
  gmx mdrun -deffnm nvt_11 -nb gpu -pme gpu

Reading file nvt_11.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      647.430      107.906      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       80.071        0.300        2.158            3.007

GROMACS reminds you: "People are DNA's way of making more DNA." (Edward O. Wilson)

State 11 complete. Cooling down fanless chassis for 5 minutes (300s)...
=========================================
RUNNING NVT STATE 12 (GPU Accelerated)
=========================================
```

```
                    :-) GROMACS - gmx mdrun, 2026.3 (-:

Executable:   /opt/gromacs/bin/gmx
Data prefix:  /opt/gromacs
Working dir:  /Users/saichaitanyajagarlapudi/GROMACS_Workspace/free energy of hydration of ethanol in water calculation
Command line:
  gmx mdrun -deffnm nvt_12 -nb gpu -pme gpu

Reading file nvt_12.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      638.063      106.345      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       81.247        0.295        2.127            3.051

GROMACS reminds you: "The scientist is not the person who always gives the right answers, he is the one who asks the right questions." (Claude Levi-Strauss)

State 12 complete. Cooling down fanless chassis for 5 minutes (300s)...
```

```
Executable:   /opt/gromacs/bin/gmx
Data prefix:  /opt/gromacs
Working dir:  /Users/saichaitanyajagarlapudi/GROMACS_Workspace/free energy of hydration of ethanol in water calculation
Command line:
  gmx mdrun -deffnm nvt_13 -nb gpu -pme gpu

Reading file nvt_13.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      636.770      106.129      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       81.412        0.295        2.123            3.057

GROMACS reminds you: "Can't You Make This Thing Go Faster ?" (Black Crowes)

State 13 complete. Cooling down fanless chassis for 5 minutes (300s)...
========================================
RUNNING NVT STATE 14 (GPU Accelerated)
```

```
Executable:   /opt/gromacs/bin/gmx
Data prefix:  /opt/gromacs
Working dir:  /Users/saichaitanyajagarlapudi/GROMACS_Workspace/free energy of hydration of ethanol in water calculation
Command line:
  gmx mdrun -deffnm nvt_14 -nb gpu -pme gpu

Reading file nvt_14.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     6906.347     1151.059      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:        7.506        3.197       23.021            0.282

GROMACS reminds you: "I've basically become a vegetarian since the only meat I'm eating is from animals I've killed myself" (Mark Zuckerberg)

State 14 complete. Cooling down fanless chassis for 5 minutes (300s)...
^Z
zsh: suspended  ./run_nvt.sh
```

The last run gave 7 ns/day because the laptop went in sleep mode due to inactivity so until I woke it up it wasn't doing anything, It's not a hardware or lambda issue.

```
for i in {15..20}; do
    echo "========================================"
    echo "RUNNING NVT STATE $i (GPU Accelerated)"
    echo "========================================"
    gmx grompp -f nvt_$i.mdp -c em_$i.gro -p topol.top -o nvt_$i.tpr -maxwarn 2
    gmx mdrun -deffnm nvt_$i -nb gpu -pme gpu
    echo "State $i complete. Cooling down fanless chassis for 5 minutes (300s)..."
    sleep 300
done
```

```
Command line:
  gmx mdrun -deffnm nvt_15 -nb gpu -pme gpu

Reading file nvt_15.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     1346.788      224.467      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       38.492        0.624        4.489            1.445

GROMACS reminds you: "I admired Bohr very much. We had long talks together, long talks in which Bohr did practically all the talking." (Paul Dirac)

State 15 complete. Cooling down fanless chassis for 5 minutes (300s)...
```

```
Command line:
  gmx mdrun -deffnm nvt_16 -nb gpu -pme gpu

Reading file nvt_16.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      663.379      110.564      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       78.146        0.307        2.211            2.935

GROMACS reminds you: "Let Me Do This" (Urban Dance Squad)

State 16 complete. Cooling down fanless chassis for 5 minutes (300s)...
```

```
Command line:
  gmx mdrun -deffnm nvt_17 -nb gpu -pme gpu

Reading file nvt_17.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      666.886      111.149      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       77.735        0.309        2.223            2.919

GROMACS reminds you: "Chemical gases filling lungs of little ones" (Black Eyed Peas)

State 17 complete. Cooling down fanless chassis for 5 minutes (300s)...
```

```
  gmx mdrun -deffnm nvt_18 -nb gpu -pme gpu

Reading file nvt_18.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      663.891      110.649      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       78.086        0.307        2.213            2.932

GROMACS reminds you: "The only place success comes before work is in the dictionary" (Vince Lombardi)

State 18 complete. Cooling down fanless chassis for 5 minutes (300s)...
=======================================
```

```
  gmx mdrun -deffnm nvt_19 -nb gpu -pme gpu

Reading file nvt_19.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      668.283      111.381      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       77.573        0.309        2.228            2.913

GROMACS reminds you: "I Live the Life They Wish They Did" (Tricky)

State 19 complete. Cooling down fanless chassis for 5 minutes (300s)...
=======================================
```

```
Command line:
  gmx mdrun -deffnm nvt_20 -nb gpu -pme gpu

Reading file nvt_20.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      667.156      111.193      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       77.704        0.309        2.224            2.918

GROMACS reminds you: "The best way to obtain plausible negative examples is to run a docking program with a biophysics-based function." (Julie Bernauer)

State 20 complete. Cooling down fanless chassis for 5 minutes (300s)...
```

```
for i in {0..4}; do
    echo "=========================================="
```

```
    echo "RUNNING NPT STATE $i (GPU Accelerated)"
    echo "========================================"
    gmx grompp -f npt_$i.mdp -c nvt_$i.gro -t nvt_$i.cpt -p topol.top -o npt_$i.tpr
-maxwarn 2
    gmx mdrun -deffnm npt_$i -nb gpu -pme gpu
    echo "State $i complete. Cooling down fanless chassis for 5 minutes..."
    sleep 300
done
```

```
  gmx mdrun -deffnm npt_2 -nb gpu -pme gpu

Reading file npt_2.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      790.940      131.824      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       65.543        0.366        2.636            2.461

GROMACS reminds you: "I don't fear death because I don't fear anything I don't understand." (Hedy Lamarr)
```

```
  gmx mdrun -deffnm npt_3 -nb gpu -pme gpu

Reading file npt_3.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      737.924      122.988      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       70.252        0.342        2.460            2.638

GROMACS reminds you: "You Can Be Too Early, You Can Be Too Late and You Can Be On Time" (J. Cruijff)
```

```
Working dir:  /Users/saichaitanyajagarlapudi/GROMACS_Workspace/free energy of hydration of ethanol in water calcula
Command line:
  gmx mdrun -deffnm npt_4 -nb gpu -pme gpu

Reading file npt_4.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      751.786      125.299      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       68.957        0.348        2.506            2.589
```

```
for i in {5..9}; do
    echo "=========================================="
    echo "RUNNING NPT STATE $i (GPU Accelerated)"
    echo "=========================================="
    gmx grompp -f npt_$i.mdp -c nvt_$i.gro -t nvt_$i.cpt -p topol.top -o npt_$i.tpr -maxwarn 2
    gmx mdrun -deffnm npt_$i -nb gpu -pme gpu
    echo "State $i complete. Cooling down fanless chassis for 5 minutes..."
```

```
    sleep 300
done
```

```
  gmx mdrun -deffnm npt_5 -nb gpu -pme gpu

Reading file npt_5.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      700.550      116.759      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       74.000        0.324        2.335            2.779

GROMACS reminds you: "Don't Push Me, Cause I'm Close to the Edge" (Grandmaster Flash)
```

```
Command line:
  gmx mdrun -deffnm npt_6 -nb gpu -pme gpu

Reading file npt_6.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      727.605      121.269      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       71.248        0.337        2.425            2.676

GROMACS reminds you: "Never attribute to malice that which can be adequately explained by stupidity." (Robert Hanlo
```

```
1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      698.816      116.470      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       74.183        0.324        2.329            2.786

GROMACS reminds you: "Is This the Right Room for an Argument ?" (Monty Python)

State 7 complete. Cooling down fanless chassis for 5 minutes...
=======================================
RUNNING NPT STATE 8 (GPU Accelerated)
=======================================
                     :-) GROMACS - gmx grompp, 2026.3 (-:
```

```
Working dir:  /Users/saichaitanyajagarlapudi/GROMACS_Workspace/free energy of hydration of ethanol in water calcula
Command line:
  gmx mdrun -deffnm npt_8 -nb gpu -pme gpu

Reading file npt_8.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      701.401      116.901      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       73.910        0.325        2.338            2.775
```

```
Reading file npt_9.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      706.017      117.670      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       73.427        0.327        2.353            2.757

GROMACS reminds you: "When you get right down to it, almost every explanation Man came up with for anything until a
1926 was stupid." (Dave Barry)

State 9 complete. Cooling down fanless chassis for 5 minutes...
```

```
for i in {10..14}; do
    echo "========================================="
    echo "RUNNING NPT STATE $i (GPU Accelerated)"
    echo "========================================="
    gmx grompp -f npt_$i.mdp -c nvt_$i.gro -t nvt_$i.cpt -p topol.top -o npt_$i.tpr
-maxwarn 2
    gmx mdrun -deffnm npt_$i -nb gpu -pme gpu
    echo "State $i complete. Cooling down fanless chassis for 5 minutes..."
    sleep 200
done
```

```
1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      698.169      116.362      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       74.252        0.323        2.327            2.788

GROMACS reminds you: "By denying scientific principles, one may maintain any paradox." (Galileo Galilei)

State 10 complete. Cooling down fanless chassis for 5 minutes...
========================================
RUNNING NPT STATE 11 (GPU Accelerated)
========================================
                     :-) GROMACS - gmx grompp, 2026.3 (-:

Executable:   /opt/gromacs/bin/gmx
```

```
Reading file npt_11.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      720.114      120.020      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       71.990        0.333        2.400            2.703

GROMACS reminds you: "Life in the streets is not easy" (Marky Mark)

State 11 complete. Cooling down fanless chassis for 5 minutes...
```

```
Command line:
  gmx mdrun -deffnm npt_12 -nb gpu -pme gpu

Reading file npt_12.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      758.916      126.487      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       68.309        0.351        2.530            2.565

GROMACS reminds you: "Schrödinger's backup: The condition of any backup is unknown until a restore is attempted." (A
```

```
  gmx mdrun -deffnm npt_13 -nb gpu -pme gpu

Reading file npt_13.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      855.620      142.604      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       60.588        0.396        2.852            2.275

GROMACS reminds you: "I ought to warn you, I have no faith" (Jane Eyre in Jane Eyre by Charlotte Bronte)
```

```
Working dir:  /Users/saichaitanyajagarlapudi/GROMACS_Workspace/free energy of hydration of ethanol in water calculation
Command line:
  gmx mdrun -deffnm npt_14 -nb gpu -pme gpu

Reading file npt_14.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      720.391      120.066      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       71.962        0.334        2.401            2.702

GROMACS reminds you: "Same sex marriage is not a gay privilege, it's equal rights. Privilege would be something like gay
 people not paying taxes. Like churches don't." (Ricky Gervais)
```

```
for i in {15..20}; do
    echo "========================================="
    echo "RUNNING NPT STATE $i (GPU Accelerated)"
    echo "========================================="
    gmx grompp -f npt_$i.mdp -c nvt_$i.gro -t nvt_$i.cpt -p topol.top -o npt_$i.tpr -maxwarn 2
    gmx mdrun -deffnm npt_$i -nb gpu -pme gpu
    echo "State $i complete. Cooling down fanless chassis for 200 seconds…”
    sleep 200
done
```

```
Working dir:  /Users/saichaitanyajagarlapudi/GROMACS_Workspace/free energy of hydration of ethanol in water calculation
Command line:
  gmx mdrun -deffnm npt_15 -nb gpu -pme gpu

Reading file npt_15.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      707.799      117.967      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       73.242        0.328        2.359            2.750

GROMACS reminds you: "It's Not Your Fault" (Pulp Fiction)
```

```
  gmx mdrun -deffnm npt_16 -nb gpu -pme gpu

Reading file npt_16.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      692.297      115.384      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       74.882        0.321        2.308            2.812

GROMACS reminds you: "Your Country Raised You, Your Country Fed You, and Just Like Any Other Country it Will Break You"
(Gogol Bordello)

====================================
```

```
Working dir:  /Users/saichaitanyajagarlapudi/GROMACS_Workspace/free energy of hydration of ethanol in water calculation
Command line:
  gmx mdrun -deffnm npt_17 -nb gpu -pme gpu

Reading file npt_17.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      700.374      116.730      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       74.018        0.324        2.335            2.780

GROMACS reminds you: "We are perhaps not far removed from the time when we shall be able to submit the bulk of chemical
phenomena to calculation." (Joseph Gay-Lussac, 1808)
```

```
  gmx mdrun -deffnm npt_18 -nb gpu -pme gpu

Reading file npt_18.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      694.319      115.721      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       74.664        0.321        2.314            2.804

GROMACS reminds you: "I'm no model lady. A model's just an imitation of the real thing." (Mae West)

=======================================
RUNNING NPT STATE 19 (GPU Accelerated)
```

```
Command line:
  gmx mdrun -deffnm npt_19 -nb gpu -pme gpu

Reading file npt_19.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      879.511      146.586      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       58.943        0.407        2.932            2.213

GROMACS reminds you: "It seemed a good idea at first" (Gerrit Groenhof)
```

```
Command line:
  gmx mdrun -deffnm npt_20 -nb gpu -pme gpu

Reading file npt_20.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      675.642      112.608      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       76.728        0.313        2.252            2.881
```

```
for i in {0..10}; do
   echo "======================================================"
   echo "LAUNCHING ALCHEMICAL PRODUCTION STATE: LAMBDA $i"
   echo "======================================================"

   # 1. Assemble the standalone production binary for the current window
   gmx grompp -f md_$i.mdp -c npt_$i.gro -t npt_$i.cpt -p topol.top -o md_$i.tpr
-maxwarn 2
```

```
    # 2. Compute the 500,000-step (1 ns) trajectory
    echo "Processing 1 ns alchemical integration for lambda $i..."
    gmx mdrun -deffnm md_$i -nb gpu -pme gpu

    # 3. Allow passive chassis housings to completely drop thermal load
    echo "State $i finalized. Cooling fanless chassis for 10 minutes..."
    sleep 600
done
```

```
Command line:
  gmx mdrun -deffnm md_0 -nb gpu -pme gpu


Back Off! I just backed up md_0.log to ./#md_0.log.1#
Reading file md_0.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads


Back Off! I just backed up md_0.xtc to ./#md_0.xtc.1#

Back Off! I just backed up md_0.edr to ./#md_0.edr.1#

Back Off! I just backed up md_0.xvg to ./#md_0.xvg.1#
starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     9116.230     1519.379      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       56.865        0.422        3.039            2.135

GROMACS reminds you: "Requiem, bring the dissident from slumber" (Bad Religion)
```

For shell script use chmod +x to let zsh give permission to run the shell script

```
  gmx mdrun -deffnm md_1 -nb gpu -pme gpu

Reading file md_1.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     8819.725     1469.955      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       58.777        0.408        2.940            2.207

GROMACS reminds you: "We'll celebrate a woman for anything, as long as it's not her talent." (Colleen McCullough)

State 1 finalized. Cooling fanless chassis for 10 minutes...
```

```
Command line:
  gmx mdrun -deffnm md_2 -nb gpu -pme gpu


Back Off! I just backed up md_2.log to ./#md_2.log.1#
Reading file md_2.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads


Back Off! I just backed up md_2.xtc to ./#md_2.xtc.1#

Back Off! I just backed up md_2.edr to ./#md_2.edr.1#

Back Off! I just backed up md_2.xvg to ./#md_2.xvg.1#
starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     7767.617     1294.611      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       66.738        0.360        2.589            2.506

GROMACS reminds you: "It's Calling Me to Break my Bonds, Again..." (Van der Graaf)
```

```
  gmx mdrun -deffnm md_3 -nb gpu -pme gpu

Reading file md_3.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     8447.208     1407.877      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       61.369        0.391        2.816            2.305
```

```
  gmx mdrun -deffnm md_4 -nb gpu -pme gpu

Reading file md_4.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     9108.136     1518.032      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       56.916        0.422        3.036            2.137

GROMACS reminds you: "Everything is failing" (Paul Bauer)
```

```
  gmx mdrun -deffnm md_5 -nb gpu -pme gpu

Reading file md_5.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.
'

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     9152.216     1525.370      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       56.642        0.424        3.051            2.127

GROMACS reminds you: "In physics, you don't have to go around making trouble for yourself. Nature does it for you." (Fra
nk Wilczek)
```

```
  gmx mdrun -deffnm md_6 -nb gpu -pme gpu

Reading file md_6.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     9036.792     1506.140      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       57.365        0.418        3.012            2.154

GROMACS reminds you: "What about my nose?" (Aneesur Rahman, responding to an Argonne manager arguing the long hair of Charles Bennett in his group was disreputing the lab; Retold by Michael Klein)
```

```
  gmx mdrun -deffnm md_7 -nb gpu -pme gpu

Reading file md_7.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     9789.382     1631.572      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       52.955        0.453        3.263            1.989

GROMACS reminds you: "Insane In Tha Membrane" (Cypress Hill)
```

Use pkill -f "sleep 600" or accordingly when u don’t have time and want to go to the next step in the loop in shell script

```
   gmx mdrun -deffnm md_8 -nb gpu -pme gpu

 Reading file md_8.tpr, VERSION 2026.3 (single precision)
 Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

 Update groups can not be used for this system because there are three or more consecutively coupled constraints

 1 GPU selected for this run.
 Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
   PP:0,PME:0
 PP tasks will do non-perturbed short-ranged interactions on the GPU
 PP task will update and constrain coordinates on the CPU
 PME tasks will do all aspects on the GPU
 Using 1 MPI thread
 Using 6 OpenMP threads

 starting mdrun 'Ethanol in water'
 500000 steps,   1000.0 ps.

 Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     8818.494     1469.757      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
 Performance:       58.785        0.408        2.940            2.208

 GROMACS reminds you: "Years of calculations and the stress, My science is waiting, nearly complete" (Midlake)
```

```
  gmx mdrun -deffnm md_9 -nb gpu -pme gpu


Back Off! I just backed up md_9.log to ./#md_9.log.1#
Reading file md_9.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads


Back Off! I just backed up md_9.xtc to ./#md_9.xtc.1#

Back Off! I just backed up md_9.edr to ./#md_9.edr.1#

Back Off! I just backed up md_9.xvg to ./#md_9.xvg.1#
starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

              Core t (s)   Wall t (s)        (%)
      Time:     8617.047     1436.183      600.0
                (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       60.160        0.399        2.872            2.259

GROMACS reminds you: "It's an easy game, just don't let the ball past!" (Szilard Pall)
```

tail -f md.log

```
  gmx mdrun -deffnm md_10 -nb gpu -pme gpu

Reading file md_10.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     9273.877     1545.655      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       55.899        0.429        3.091           2.099

GROMACS reminds you: "We'll celebrate a woman for anything, as long as it's not her talent." (Colleen McCullough)
```

11-12 values were given but I closed the terminal

```
Command line:
  gmx mdrun -deffnm md_13 -nb gpu -pme gpu


Back Off! I just backed up md_13.log to ./#md_13.log.1#
Reading file md_13.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads


Back Off! I just backed up md_13.xtc to ./#md_13.xtc.1#

Back Off! I just backed up md_13.edr to ./#md_13.edr.1#

Back Off! I just backed up md_13.xvg to ./#md_13.xvg.1#
starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     8603.334     1433.897      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       60.255        0.398        2.868           2.263

GROMACS reminds you: "Science Won't Change You" (The Talking Heads)
```

```
  gmx mdrun -deffnm md_14 -nb gpu -pme gpu

Reading file md_14.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     9486.214     1581.037      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       54.648        0.439        3.162            2.052

GROMACS reminds you: "And I Will Strike Down Upon Thee With Great Vengeance and With Furious Anger Those Who Attempt to
Poison and Destroy My Brothers." (Pulp Fiction)
```

```
Command line:
  gmx mdrun -deffnm md_15 -nb gpu -pme gpu

Reading file md_15.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:    10544.278     1757.381      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       49.164        0.488        3.515            1.846

GROMACS reminds you: "Oh My God ! It's the Funky Shit" (Beastie Boys)
```

```
Command line:
  gmx mdrun -deffnm md_16 -nb gpu -pme gpu

Reading file md_16.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     9638.904     1606.491      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       53.782        0.446        3.213            2.020

GROMACS reminds you: "The greatest shortcoming of the human race is our inability to understand the exponential function
." (Albert Bartlett)
```

```
Command line:
  gmx mdrun -deffnm md_17 -nb gpu -pme gpu

Reading file md_17.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     8851.364     1475.236      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       58.567        0.410        2.950            2.199
```

```
Command line:
  gmx mdrun -deffnm md_18 -nb gpu -pme gpu

Reading file md_18.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     8944.260     1490.718      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       57.959        0.414        2.981            2.176

GROMACS reminds you: "Teaching quantum computing is like teaching computer science at Hogwarts." (Thomas Sterling, ISC20
18 keynote)
```

```
Command line:
  gmx mdrun -deffnm md_19 -nb gpu -pme gpu

Reading file md_19.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     9745.455     1624.252      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       53.194        0.451        3.248            1.998

GROMACS reminds you: "I Don't Like Dirt" (The Breeders)
```

20 has been done but hasn’t been included since i closed the terminal

gmx bar -f dhdl_*.xvg -o -oi -oh

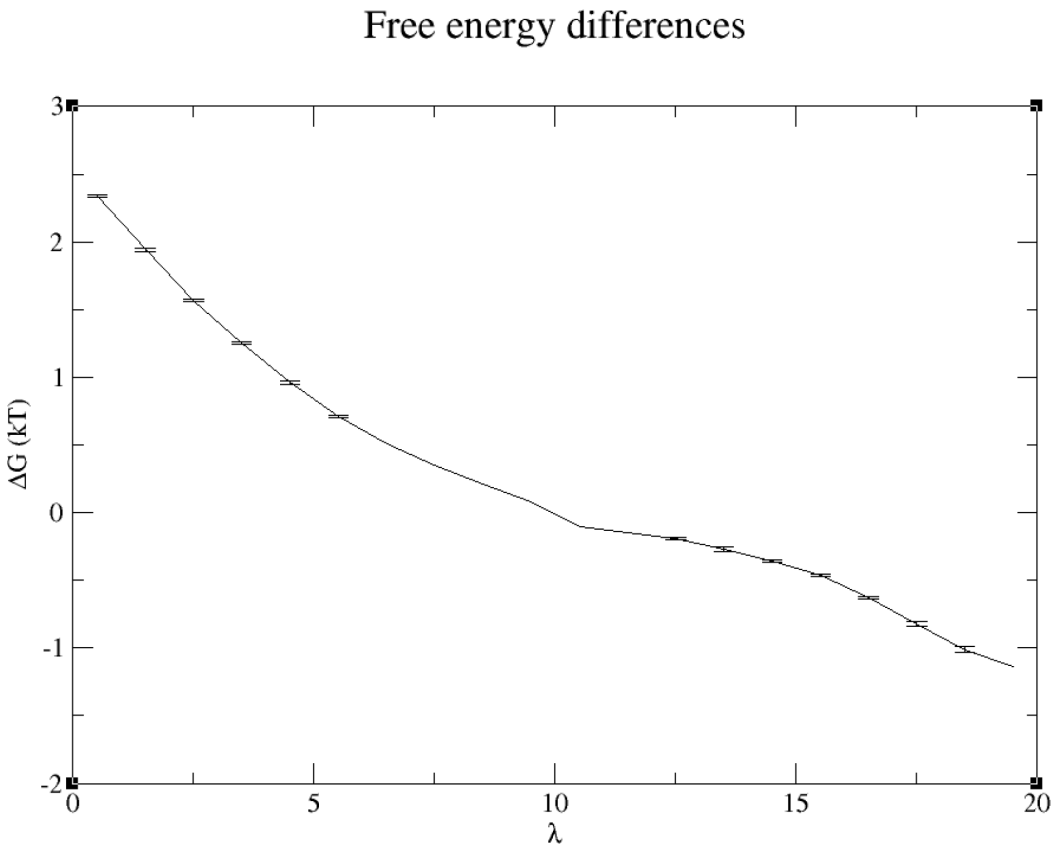
```
Final results in kJ/mol:

point      0 -      1,   DG  5.80 +/-  0.03
point      1 -      2,   DG  4.82 +/-  0.01
point      2 -      3,   DG  3.89 +/-  0.03
point      3 -      4,   DG  3.09 +/-  0.02
point      4 -      5,   DG  2.39 +/-  0.02
point      5 -      6,   DG  1.77 +/-  0.01
point      6 -      7,   DG  1.27 +/-  0.01
point      7 -      8,   DG  0.86 +/-  0.01
point      8 -      9,   DG  0.51 +/-  0.01
point      9 -     10,   DG  0.19 +/-  0.01
point     10 -     11,   DG -0.25 +/-  0.01
point     11 -     12,   DG -0.37 +/-  0.01
point     12 -     13,   DG -0.48 +/-  0.02
point     13 -     14,   DG -0.66 +/-  0.04
point     14 -     15,   DG -0.89 +/-  0.02
point     15 -     16,   DG -1.15 +/-  0.03
point     16 -     17,   DG -1.57 +/-  0.03
point     17 -     18,   DG -2.03 +/-  0.06
point     18 -     19,   DG -2.51 +/-  0.05
point     19 -     20,   DG -2.83 +/-  0.01

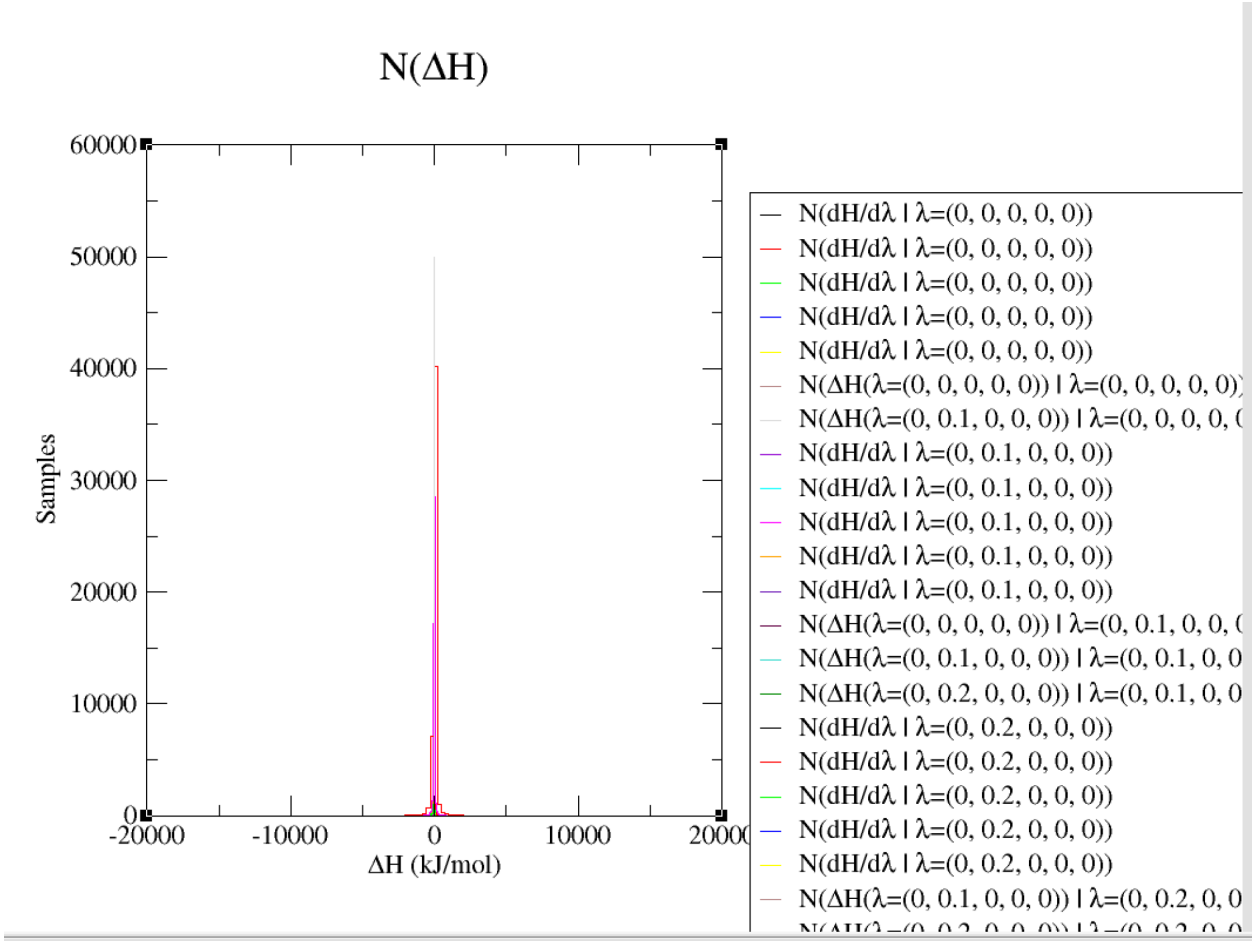
total      0 -     20,   DG 11.86 +/-  0.19


GROMACS reminds you: "Jesus Not Only Saves, He Also Frequently Makes Backups." (Myron Bradshaw)
```

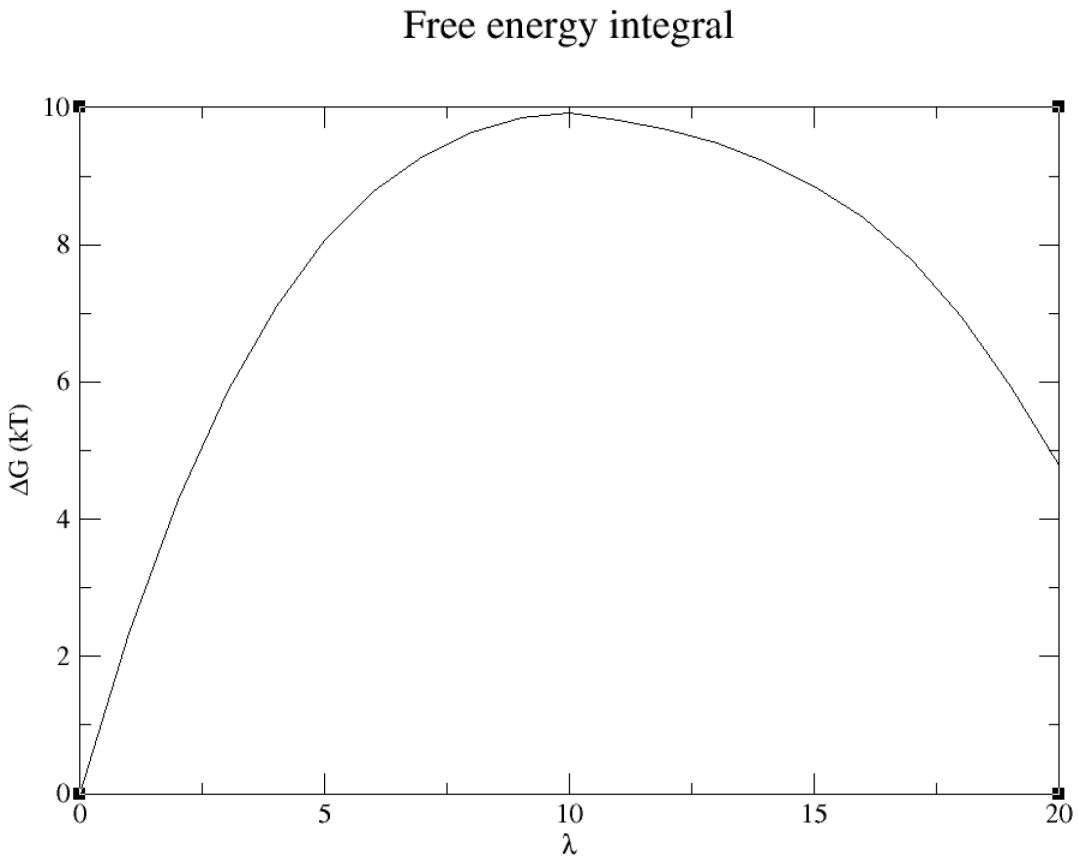
xmgrace bar.xvg



xmgrace histogram.xvg



xmgrace barint.xvg



Sorry, the full scheme hasn't been done, the perl command gave only 20 mdp files but we were supposed to do 30 simulations

```
for i in $(seq 21 30); do
   echo "=== Running EM for state $i ==="

   gmx grompp -f em_$i.mdp -c npt_20.gro -p topol.top -o em_$i.tpr -maxwarn 1
   # If you have a different starting structure, change npt_20.gro to the
appropriate .gro file

   gmx mdrun -v -deffnm em_$i
done
for i in {21..30}; do
   echo "==========================================="
   echo "RUNNING NVT STATE $i (GPU Accelerated)"
   echo "==========================================="
```

```
    gmx grompp -f nvt_$i.mdp -c em_$i.gro -p topol.top -o nvt_$i.tpr -maxwarn 2
    gmx mdrun -deffnm nvt_$i -nb gpu -pme gpu
    echo "State $i complete. Cooling down fanless chassis for 5 minutes (300s)..."
    sleep 300
done
```

```
   gmx mdrun -deffnm nvt_21 -nb gpu -pme gpu

Reading file nvt_21.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      746.129      124.356      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       69.479        0.345        2.487            2.609

GROMACS reminds you: "Drugs are Bad, mmokay" (South Park)

State 21 complete. Cooling down fanless chassis for 5 minutes (300s)...
```

```
Command line:
  gmx mdrun -deffnm nvt_22 -nb gpu -pme gpu

Reading file nvt_22.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      797.958      132.994      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       64.966        0.369        2.660            2.440

GROMACS reminds you: "Raw data is like raw sewage, it requires some processing before it can be spread around. The opposite is true of theories." (Jim Carr)
```

```
Command line:
  gmx mdrun -deffnm nvt_23 -nb gpu -pme gpu

Reading file nvt_23.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     1113.065      185.512      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       46.575        0.515        3.710            1.749

GROMACS reminds you: "We ignore public understanding of science at our peril." (Eugenie Clark)
```

```
Command line:
  gmx mdrun -deffnm nvt_24 -nb gpu -pme gpu

Reading file nvt_24.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      943.559      157.266      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       54.940        0.437        3.145            2.063

GROMACS reminds you: "Ich Bin Ein Berliner" (J.F. Kennedy)
```

```
Command line:
  gmx mdrun -deffnm nvt_25 -nb gpu -pme gpu

Reading file nvt_25.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      758.115      126.361      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       68.377        0.351        2.527            2.568

GROMACS reminds you: "I always seem to get inspiration and renewed vitality by contact with this great novel land of yours which sticks up out of the Atlantic." (Winston Churchill)
```

```
Command line:
  gmx mdrun -deffnm nvt_26 -nb gpu -pme gpu

Reading file nvt_26.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     1881.093      313.525      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       27.558        0.871        6.270           1.035

GROMACS reminds you: "Expertise is not inherently good." (Joe Jordan)
```

```
Command line:
  gmx mdrun -deffnm nvt_27 -nb gpu -pme gpu

Reading file nvt_27.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     1345.485      224.249      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       38.529        0.623        4.485           1.447

GROMACS reminds you: "There are only two hard things in computer science – cache invalidation, naming things and off-by-one errors." (Anonymous)
```

```
Command line:
  gmx mdrun -deffnm nvt_28 -nb gpu -pme gpu

Reading file nvt_28.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      808.584      134.765      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       64.113        0.374        2.695           2.408

GROMACS reminds you: "The Poodle Bites" (F. Zappa)
```

```
Command line:
  gmx mdrun -deffnm nvt_29 -nb gpu -pme gpu

Reading file nvt_29.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      945.219      157.544      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       54.843        0.438        3.151           2.059

GROMACS reminds you: "Right Between the Eyes" (F. Zappa)
```

```
Command line:
  gmx mdrun -deffnm nvt_30 -nb gpu -pme gpu

Reading file nvt_30.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     1045.848      174.309      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       49.568        0.484        3.486            1.861

GROMACS reminds you: "Science, for me, gives a partial explanation for life. In so far as it goes, it is based on fact, experience and experiment." (Rosalind Franklin)
```

```
for i in {21..30}; do
    echo "==========================================="
    echo "RUNNING NPT STATE $i (GPU Accelerated)"
    echo "==========================================="
    gmx grompp -f npt_$i.mdp -c nvt_$i.gro -t nvt_$i.cpt -p topol.top -o npt_$i.tpr
-maxwarn 2
    gmx mdrun -deffnm npt_$i -nb gpu -pme gpu
    echo "State $i complete. Cooling down fanless chassis for 5 minutes..."
    sleep 300
done
```

```
Command line:
  gmx mdrun -deffnm npt_21 -nb gpu -pme gpu

Reading file npt_21.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     783.498      130.584      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       66.166        0.363        2.612            2.485

GROMACS reminds you: "We haven't the money, so we've got to think." (Ernest Rutherford)
```

```
Command line:
  gmx mdrun -deffnm npt_22 -nb gpu -pme gpu

Reading file npt_22.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     1136.763      189.462      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       45.604        0.526        3.789            1.713

GROMACS reminds you: "What's the point, yo, what's the spread?" (Red Hot Chili Peppers)
```

```
Command line:
  gmx mdrun -deffnm npt_23 -nb gpu -pme gpu

Reading file npt_23.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      718.548      119.760      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       72.146        0.333        2.395            2.709

GROMACS reminds you: "You Can Always Go On Ricky Lake" (Offspring)
```

```
Command line:
  gmx mdrun -deffnm npt_24 -nb gpu -pme gpu

Reading file npt_24.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      790.769      131.796      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       65.557        0.366        2.636            2.462

GROMACS reminds you: "I'm not interrupting you, I'm putting our conversation in full-duplex mode." (Antone Roundy)
```

```
Command line:
  gmx mdrun -deffnm npt_25 -nb gpu -pme gpu

Reading file npt_25.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      697.928      116.322      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       74.278        0.323        2.326            2.789

GROMACS reminds you: "If mathematical analysis should ever hold a prominent place in chemistry - an aberration which is happily almost impossible - it would occasion a rapid and widespread degeneration of that science." (Aguste
Comte, 1830)
```

```
  gmx mdrun -deffnm npt_26 -nb gpu -pme gpu

Reading file npt_26.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      773.817      128.971      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       66.993        0.358        2.579            2.516

GROMACS reminds you: "They Were So Quiet About It" (Pixies)

State 26 complete. Cooling down fanless chassis for 5 minutes...
```

```
Command line:
  gmx mdrun -deffnm npt_27 -nb gpu -pme gpu

Reading file npt_27.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      754.025      125.672      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       68.752        0.349        2.513            2.582

GROMACS reminds you: "I think it would be a good idea." (Mahatma Gandhi, when asked what he thought of Western civilization)
```

```
  gmx mdrun -deffnm npt_28 -nb gpu -pme gpu

Reading file npt_28.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      729.730      121.623      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       71.041        0.338        2.432            2.668

GROMACS reminds you: "It Just Tastes Better" (Burger King)

State 28 complete. Cooling down fanless chassis for 5 minutes...
```

```
Command line:
  gmx mdrun -deffnm npt_29 -nb gpu -pme gpu

Reading file npt_29.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:      853.663      142.278      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       60.727        0.395        2.846            2.280

GROMACS reminds you: "He's using code that only you and I know" (Kate Bush)
```

```
  gmx mdrun -deffnm npt_30 -nb gpu -pme gpu

Reading file npt_30.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
50000 steps,    100.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     1286.965      214.502      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       40.280        0.596        4.290           1.513

GROMACS reminds you: "Why is it that programmers always confuse Halloween with Christmas? Because 31 OCT = 25 DEC." (Anonymous)

State 30 complete. Cooling down fanless chassis for 5 minutes...
```

```
#!/bin/bash
for i in {21..25}; do
   echo "============================================================"
   echo "LAUNCHING ALCHEMICAL PRODUCTION STATE: LAMBDA $i"
   echo "============================================================"

   # 1. Assemble the standalone production binary for the current window
   gmx grompp -f md_$i.mdp -c npt_$i.gro -t npt_$i.cpt -p topol.top -o md_$i.tpr -maxwarn 2

   # 2. Compute the 500,000-step (1 ns) trajectory
   echo "Processing 1 ns alchemical integration for lambda $i..."
   gmx mdrun -deffnm md_$i -nb gpu -pme gpu

   # 3. Allow passive chassis housings to completely drop thermal load
   echo "State $i finalized. Cooling fanless chassis for 10 minutes..."
   sleep 600
done
```

```
Command line:
  gmx mdrun -deffnm md_21 -nb gpu -pme gpu

Reading file md_21.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     9432.243     1572.051      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       54.960        0.437        3.144           2.064

GROMACS reminds you: "I'll Master Your Language, and In the Meantime I'll Create My Own" (Tricky)
```

```
Command line:
  gmx mdrun -deffnm md_22 -nb gpu -pme gpu

Reading file md_22.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     8357.068     1392.852      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       62.031        0.387        2.786            2.329

GROMACS reminds you: "Live for Liposuction" (Robbie Williams)
```

```
Command line:
  gmx mdrun -deffnm md_23 -nb gpu -pme gpu


Back Off! I just backed up md_23.log to ./#md_23.log.1#
Reading file md_23.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads


Back Off! I just backed up md_23.xtc to ./#md_23.xtc.1#

Back Off! I just backed up md_23.edr to ./#md_23.edr.1#

Back Off! I just backed up md_23.xvg to ./#md_23.xvg.1#
starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     7551.152     1258.526      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       68.652        0.350        2.517            2.578

GROMACS reminds you: "All Beauty Must Die" (Nick Cave)
```

```
  gmx mdrun -deffnm md_24 -nb gpu -pme gpu

Reading file md_24.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.343

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     8411.169     1401.869      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       61.632        0.389        2.804            2.314

GROMACS reminds you: "Since the mathematicians have invaded the theory of relativity I do not understand it myself any m
ore." (Albert Einstein)
```

```
Command line:
  gmx mdrun -deffnm md_25 -nb gpu -pme gpu

Reading file md_25.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     8590.377     1431.737      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       60.346        0.398        2.863            2.266

GROMACS reminds you: "Jesus Built My Hotrod" (Ministry)
```

```
Executable:   /opt/gromacs/bin/gmx
Data prefix:  /opt/gromacs
Working dir:  /Users/saichaitanyajagarlapudi/GROMACS_Workspace/free energy of hydration of ethanol in water calculation
Command line:
  gmx mdrun -deffnm md_26 -nb gpu -pme gpu

Reading file md_26.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads

starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

               Core t (s)   Wall t (s)        (%)
       Time:     9160.874     1526.819      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       56.588        0.424        3.054            2.125

GROMACS reminds you: "Martin [Karplus] had a green laser, Arieh [Warshel] had a red laser, I have a *blue* laser" (Michael Levitt, Nob
el lecture 2013)
```

```
  gmx mdrun -deffnm md_28 -nb gpu -pme gpu


Back Off! I just backed up md_28.log to ./#md_28.log.1#
Reading file md_28.tpr, VERSION 2026.3 (single precision)
Changing nstlist from 20 to 100, rlist from 1.2 to 1.342

Update groups can not be used for this system because there are three or more consecutively coupled constraints

1 GPU selected for this run.
Mapping of GPU IDs to the 2 GPU tasks in the 1 rank on this node:
  PP:0,PME:0
PP tasks will do non-perturbed short-ranged interactions on the GPU
PP task will update and constrain coordinates on the CPU
PME tasks will do all aspects on the GPU
Using 1 MPI thread
Using 6 OpenMP threads


Back Off! I just backed up md_28.xtc to ./#md_28.xtc.1#

Back Off! I just backed up md_28.edr to ./#md_28.edr.1#

Back Off! I just backed up md_28.xvg to ./#md_28.xvg.1#
starting mdrun 'Ethanol in water'
500000 steps,   1000.0 ps.

Writing final coordinates.

             Core t (s)   Wall t (s)        (%)
       Time:     9917.427     1652.906      600.0
                 (ns/day)    (hour/ns)    (ms/step)  (Matom*steps/s)
Performance:       52.272        0.459        3.306            1.963

GROMACS reminds you: "If at one time or another I have brushed a few colleagues the wrong way, I must apologize: I had not realized that they were covered with fur." (Edwin Chargaff)

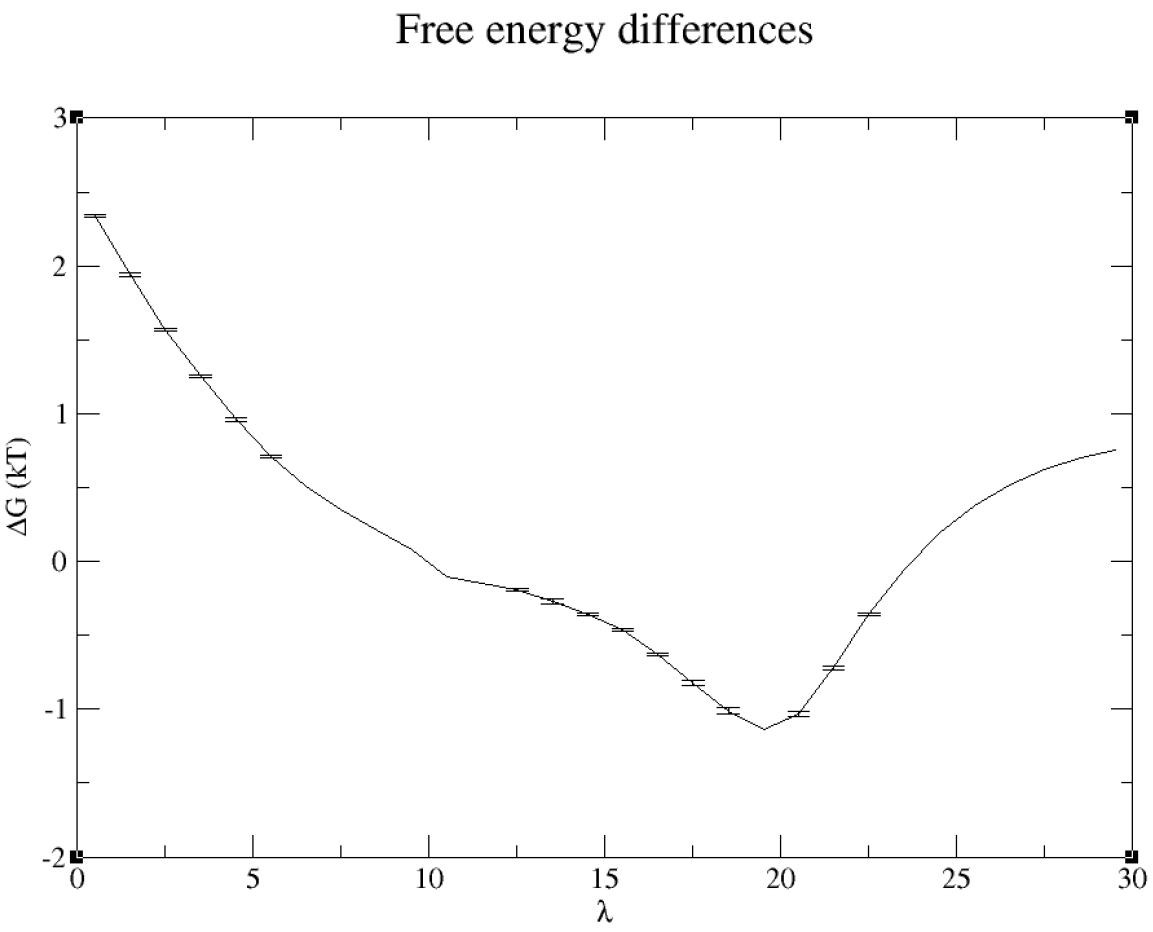
State 28 finalized. Cooling fanless chassis for 10 minutes...
```

```
Final results in kJ/mol:

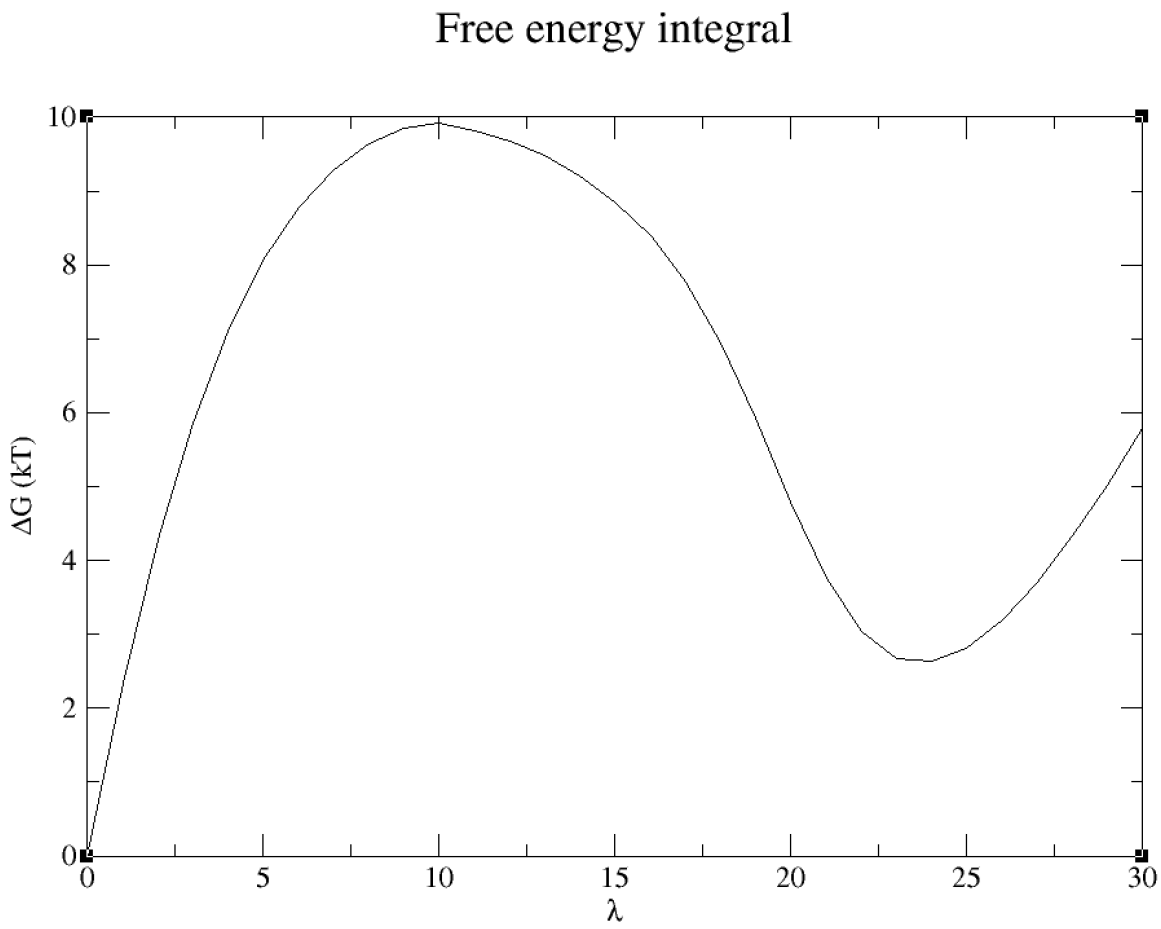
point      0 -      1,   DG  5.80 +/-  0.03
point      1 -      2,   DG  4.82 +/-  0.01
point      2 -      3,   DG  3.89 +/-  0.03
point      3 -      4,   DG  3.09 +/-  0.02
point      4 -      5,   DG  2.39 +/-  0.02
point      5 -      6,   DG  1.77 +/-  0.01
point      6 -      7,   DG  1.27 +/-  0.01
point      7 -      8,   DG  0.86 +/-  0.01
point      8 -      9,   DG  0.51 +/-  0.01
point      9 -     10,   DG  0.19 +/-  0.01
point     10 -     11,   DG -0.25 +/-  0.01
point     11 -     12,   DG -0.37 +/-  0.01
point     12 -     13,   DG -0.48 +/-  0.02
point     13 -     14,   DG -0.66 +/-  0.04
point     14 -     15,   DG -0.89 +/-  0.02
point     15 -     16,   DG -1.15 +/-  0.03
point     16 -     17,   DG -1.57 +/-  0.03
point     17 -     18,   DG -2.03 +/-  0.06
point     18 -     19,   DG -2.51 +/-  0.05
point     19 -     20,   DG -2.83 +/-  0.01
point     20 -     21,   DG -2.56 +/-  0.04
point     21 -     22,   DG -1.77 +/-  0.03
point     22 -     23,   DG -0.90 +/-  0.01
point     23 -     24,   DG -0.13 +/-  0.00
point     24 -     25,   DG  0.48 +/-  0.00
point     25 -     26,   DG  0.93 +/-  0.00
point     26 -     27,   DG  1.28 +/-  0.00
point     27 -     28,   DG  1.53 +/-  0.00
point     28 -     29,   DG  1.73 +/-  0.00
point     29 -     30,   DG  1.86 +/-  0.00

total      0 -     30,   DG 14.31 +/-  0.14
```

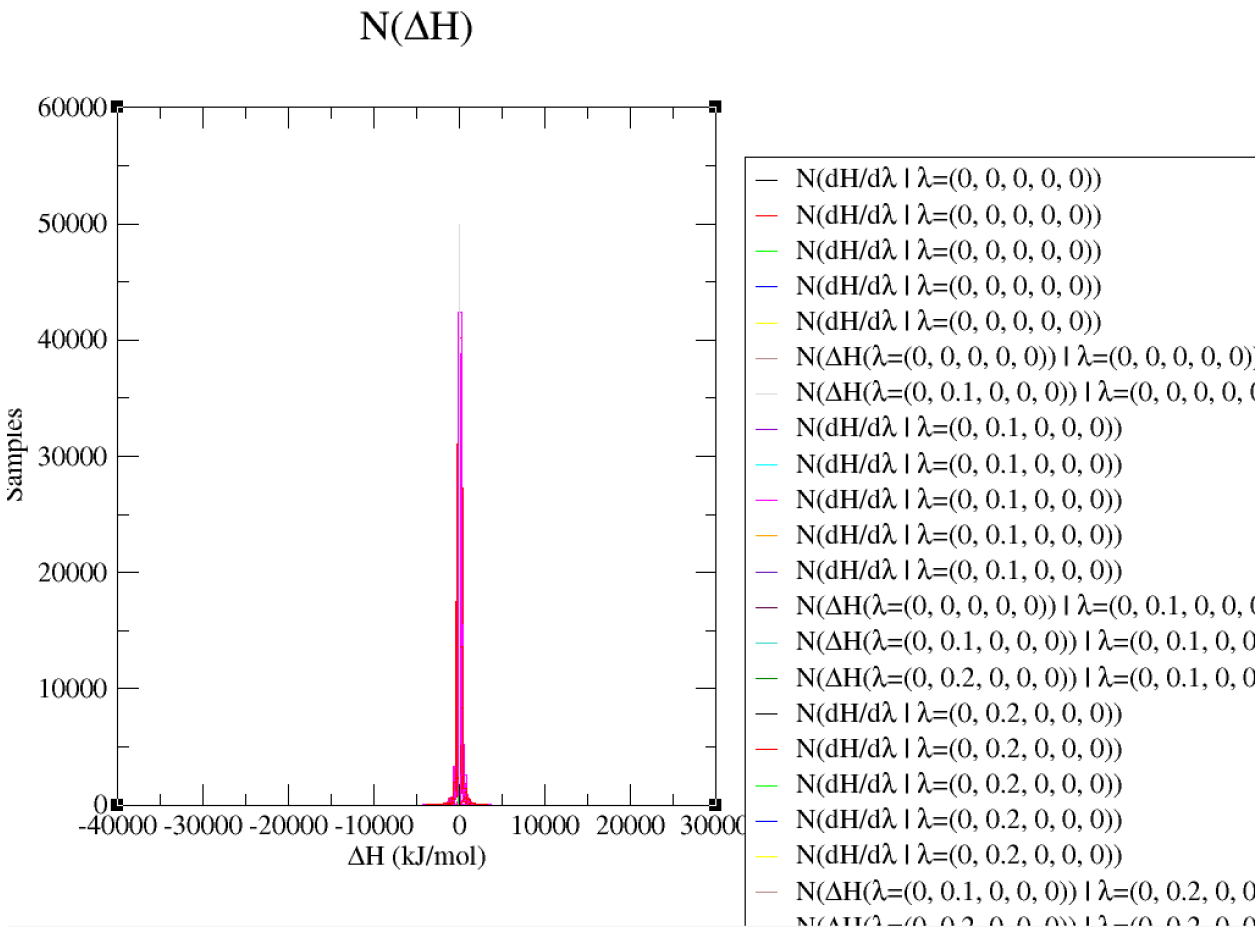
xmgrace bar.xvg



xmgrace barint.xvg



xmgrace histogram.xvg


N(ΔH)
60000
50000
40000
30000
20000
10000
0
Samples
-40000 -30000 -20000 -10000 0 10000 20000 30000
ΔH (kJ/mol)
N(dH/dλ | λ=(0, 0, 0, 0, 0))
N(dH/dλ | λ=(0, 0, 0, 0, 0))
N(dH/dλ | λ=(0, 0, 0, 0, 0))
N(dH/dλ | λ=(0, 0, 0, 0, 0))
N(dH/dλ | λ=(0, 0, 0, 0, 0))
N(ΔH(λ=(0, 0, 0, 0, 0)) | λ=(0, 0, 0, 0, 0)
N(ΔH(λ=(0, 0.1, 0, 0, 0)) | λ=(0, 0, 0, 0,
N(dH/dλ | λ=(0, 0.1, 0, 0, 0))
N(dH/dλ | λ=(0, 0.1, 0, 0, 0))
N(dH/dλ | λ=(0, 0.1, 0, 0, 0))
N(dH/dλ | λ=(0, 0.1, 0, 0, 0))
N(dH/dλ | λ=(0, 0.1, 0, 0, 0))
N(ΔH(λ=(0, 0, 0, 0, 0)) | λ=(0, 0.1, 0, 0,
N(ΔH(λ=(0, 0.1, 0, 0, 0)) | λ=(0, 0.1, 0, 0
N(ΔH(λ=(0, 0.2, 0, 0, 0)) | λ=(0, 0.1, 0, 0
N(dH/dλ | λ=(0, 0.2, 0, 0, 0))
N(dH/dλ | λ=(0, 0.2, 0, 0, 0))
N(dH/dλ | λ=(0, 0.2, 0, 0, 0))
N(dH/dλ | λ=(0, 0.2, 0, 0, 0))
N(dH/dλ | λ=(0, 0.2, 0, 0, 0))
N(ΔH(λ=(0, 0.1, 0, 0, 0)) | λ=(0, 0.2, 0, 0